Micro Lab Final Exam Sample Questions

1. Which component has built -in bluetooth or Wi-Fi modules? [1]
    a. Arduino
    b. ESP 32
    c. Raspberry pi
    d. None of the above
2. What does the GND pin on a microcontroller do? [1]
    a. Sends output data
    b. Sends digital data
    c. Acts as a reference for 0V
    d. Controls the sensor
3. You want to build a system that reads MQ2 sensor values and sends alerts to a server. Compare Arduino, ESP32, and Raspberry Pi 5 for this task, and justify which board is most appropriate in terms of cost efficiency, wireless communication, and processing power. [3]
4. Write two scenarios where you would choose an ESP32 over a Raspberry Pi. [3]
5. You are planning a home security system with a central monitoring unit and multiple sensor nodes. For which part would you use a microcontroller and for which part would you use a microprocessor? Explain your reasoning. [3]
6. The following code makes an LED blink every 500 ms. Modify the code so the LED turns ON and OFF twice before the loop restarts. [4]

```
const int ledPin = 7;
void setup() {
     pinMode(ledPin, OUTPUT);
}
void loop() {
  digitalWrite(ledPin, HIGH);
  delay(500);
  digitalWrite(ledPin, LOW);
  delay(500);
```